

EXP2_DBSCAN

April 10, 2026

```
[7]: import numpy as np
import matplotlib.pyplot as plt
from sklearn.cluster import DBSCAN, KMeans
from sklearn.preprocessing import StandardScaler
from mpl_toolkits.mplot3d import Axes3D
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[9]: np.random.seed(42)
```

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[11]: cluster1 = np.random.normal(loc=[50, 30, 20], scale=5, size=(120, 3))
cluster2 = np.random.normal(loc=[80, 60, 40], scale=5, size=(120, 3))
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[13]: anomalies = np.random.uniform(low=0, high=100, size=(30, 3))
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[15]: X = np.vstack((cluster1, cluster2, anomalies))
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[17]: scaler = StandardScaler()
X_scaled = scaler.fit_transform(X)
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[19]: eps_value = 0.5
min_samples_value = 5

dbscan = DBSCAN(eps=eps_value, min_samples=min_samples_value)
db_labels = dbscan.fit_predict(X_scaled)

# Extract anomalies (noise points)
noise_mask = db_labels == -1
noise_points = X[noise_mask]

print("DBSCAN detected anomalies:", len(noise_points))
```

DBSCAN detected anomalies: 32

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[21]: kmeans = KMeans(n_clusters=2, random_state=42, n_init=10)
km_labels = kmeans.fit_predict(X_scaled)
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[23]: plt.figure(figsize=(14, 6))
```

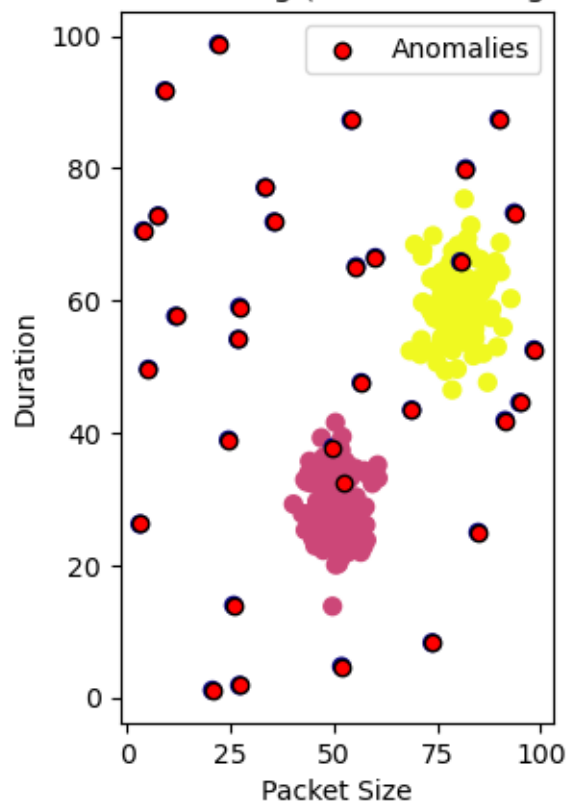
```
[23]: <Figure size 1400x600 with 0 Axes>
```

<Figure size 1400x600 with 0 Axes>

```
[25]: plt.subplot(1, 2, 1)
plt.scatter(X[:, 0], X[:, 1], c=db_labels, cmap='plasma', s=40)
plt.scatter(noise_points[:, 0], noise_points[:, 1],
            color='red', label='Anomalies', edgecolors='black')
plt.title("DBSCAN Clustering (Anomalies Highlighted)")
plt.xlabel("Packet Size")
plt.ylabel("Duration")
plt.legend()
```

[25]: <matplotlib.legend.Legend at 0x167e22030>

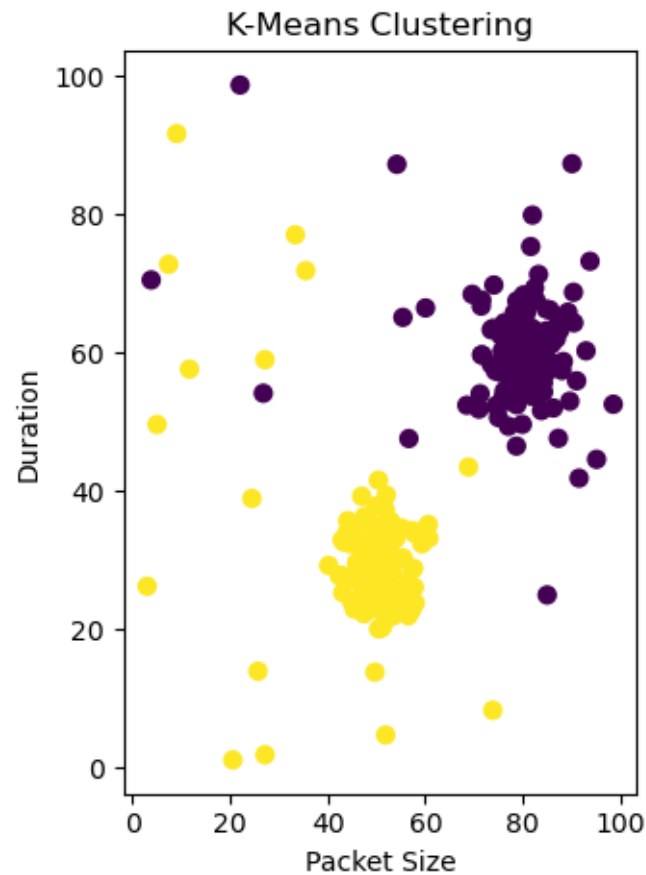
DBSCAN Clustering (Anomalies Highlighted)



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[27]: plt.subplot(1, 2, 2)
plt.scatter(X[:, 0], X[:, 1], c=km_labels, cmap='viridis', s=40)
plt.title("K-Means Clustering")
plt.xlabel("Packet Size")
plt.ylabel("Duration")

plt.tight_layout()
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plt.show()
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[29]: fig = plt.figure(figsize=(12, 6))
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<Figure size 1200x600 with 0 Axes>

```
[37]: def run_dbscan(eps, min_samples):  
    model = DBSCAN(eps=eps, min_samples=min_samples)  
    labels = model.fit_predict(X_scaled)  
    noise = np.sum(labels == -1)  
    clusters = len(set(labels)) - (1 if -1 in labels else 0)  
  
    print(f"\nFor eps={eps}, min_samples={min_samples}")  
    print("Clusters found:", clusters)  
    print("Noise points:", noise)
```

```
[39]: run_dbscan(0.3, 5)  
run_dbscan(0.5, 5)  
run_dbscan(0.7, 3)
```

For eps=0.3, min_samples=5
Clusters found: 2
Noise points: 45

For eps=0.5, min_samples=5
Clusters found: 2
Noise points: 32

For eps=0.7, min_samples=3
Clusters found: 2
Noise points: 27

```
[41]: print("DBSCAN clusters:", len(set(db_labels)) - (1 if -1 in db_labels else 0))  
      print("DBSCAN noise points:", np.sum(db_labels == -1))  
      print("K-Means clusters:", len(set(km_labels)))
```

DBSCAN clusters: 2
DBSCAN noise points: 32
K-Means clusters: 2

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